DFL Loss Landscape: Original (75 bases) vs State Aggregation (fewer bases)
Fewer LP variables ⇒ fewer basis regions ⇒ smoother landscape

State Agg (M=10, 162 bases)
State Agg (M=20, 213 bases)
State Agg (M=30, 224 bases)
$\lambda_{true} = 10.0$

DFL Loss $L(\lambda) = -r_{true}^T y^*(\lambda)$

−250
−500
−750
−1000
−1250
−1500
−1750

0
10
20
30
40
50

Gradient: state aggregation produces smoother, more informative gradients

$\partial L/\partial \lambda$
(finite diff)

200
0
−200
−400
−600
−800
−1000
−1200
−1400

DFL gradient (S=56)
State Agg gradient (M=10)
State Agg gradient (M=20)
State Agg gradient (M=30)

0
10
20
30
40
50

λ_{pred}